



Final Project Presentation:

NLP Applications in Business Healthcare

From Social Media Shortage Detection to Medical AI Agent

Yichen Wang

Ph.d. Student

Department of Information Systems & Technology Management

School of Business, The George Washington University

Healthcare Application: Challenges

Challenge 1: Early Signal Detection

- GLP-1 drug shortages affect patient access & treatment continuity
- Patient experiences embedded in noisy Reddit posts before official reports
- Need: Transform ~800K unstructured posts into shortage signals
- Approach: Fine-tuned RoBERTa encoder — binary classification

PART 1: Encoder NLP

Drug Shortage Detection from Reddit
(RoBERTa · 800K posts · F1=0.87)

Challenge 2: Medical Knowledge Access

- Physicians repeatedly answer the same refractive surgery questions
- General-purpose LLMs hallucinate clinical specifics and cite stale data
- Need: Accurate, traceable, evidence-based answers for patients
- Approach: RAG — retrieve AAO guidelines → ground LLM generation

PART 2: Medical RAG

Ophthalmology Follow-up Assistant
(DeepSeek · 800 docs · +43.4% quality)

Whole Picture

Aspect	PART 1: Shortage Detection	PART 2: Medical RAG
Input Data	800K Reddit posts (14–16 subreddits)	25 patient questions
Knowledge Base	1,200 labeled posts ($\kappa=0.85$)	800 medical docs (2.27M chars)
Method	Supervised encoder classification	Retrieval + generation (RAG)
Core Model	RoBERTa-base (fine-tuned)	BGE embeddings + DeepSeek-R1:1.5b
Output	Binary shortage signals	Natural language answers
Evaluation	F1 = 0.87 (shortage class)	Physician 5-pt rating: 4.36/5
Downstream Use	VAR model + GNN forecasting	Patient education & physician support Application

PART 1 — Drug Shortage Detection: Problem & Motivation

Context: GLP-1 Receptor Agonist Shortages (2010–2025)

Medications: Ozempic · Wegovy · Rybelsus · Victoza · Mounjaro · Zepbound | Affects patient access, treatment continuity & medication switching decisions

Traditional Data Gap

FDA reports & manufacturer announcements are lagging indicators (don't capture real-time information)

NLP Task Defined

Binary classification:
given Reddit post →
shortage-related (1)
vs. non-shortage (0)

Key Challenges

- Noisy informal text
- Class imbalance
- No-label Data

Business Question

Can social media discussions predict prescription behavior changes and serve as leading indicators?

PART 1— Dataset & Processing Pipeline

Dataset Facts

~800K

Reddit posts & comments

14–16

GLP-1 related subreddits

Jan 2010

– Jan 2025 timespan

1,200

Manually labeled posts

421

Shortage-related (label=1)

$\kappa = 0.95$

Cohen's Kappa (2 RAs + GPT)

95.6%

Overall annotation agreement

Pipeline Architecture

Reddit Collection (800K posts)

Manual Labeling (1,200 posts · $\kappa=0.95$)

RoBERTa Fine-tuning (60/20/20 split)

Full Corpus Prediction (batch inference)

LLM Filtering — o3-mini multi-layer prompting

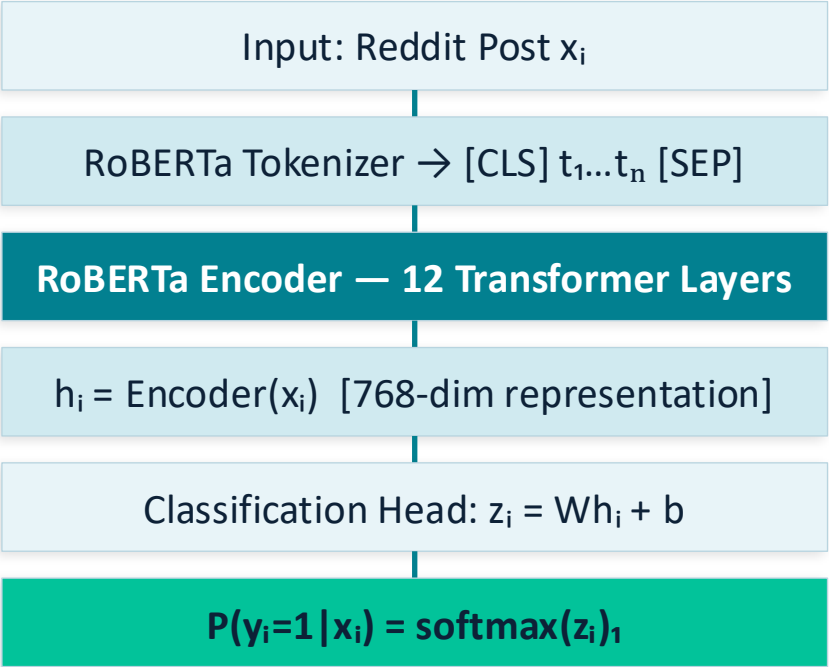
52,866 Shortage Posts Identified ✓

LDA Topic Modeling (10 topics)

Weekly Time-Series → VAR + GNN Downstream

PART 1 — Model Architecture & Hyperparameters

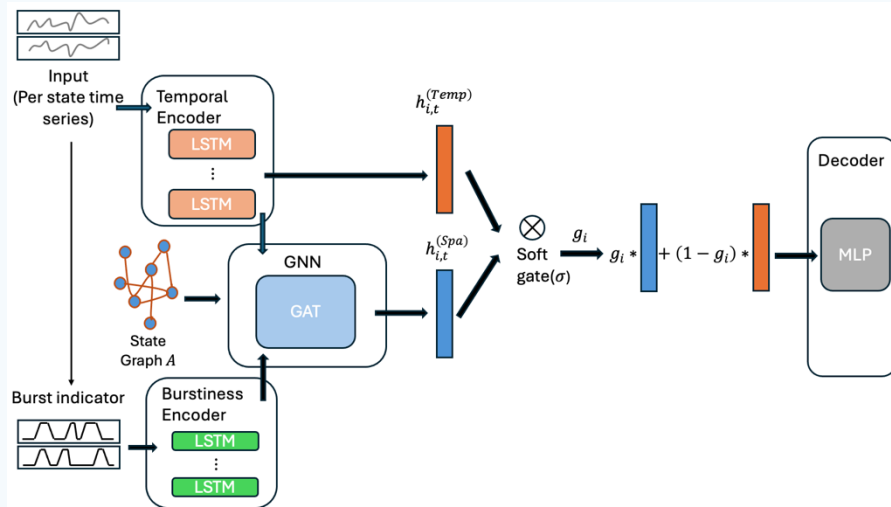
Model Architecture



Two-Stage Filtering		
800,000	Full Reddit corpus	
$\rightarrow$ 88,554	RoBERTa predictions (shortage=1)	
$\rightarrow$ 52,866	After o3-mini LLM filtering	
War	0.87	0.94
Earl	F1 Score (Shortage Class)	F1 Score (Non-Shortage Class)
Loss function	Cross-entropy	weighted CE
Layer freezing	None	Embed + layers 1–8

Next: Enhanced Causality and Prediction Power (Downstream Business Value)

Model Architecture



- Temporal Encoder (LSTM) $\rightarrow h_i^{Temp}$
- GNN $\rightarrow h_i^{Spa}$
- Burstiness Encoder (LSTM) — Burst indicator

Causality with Social Media (VAR MODEL)

$$Y_t = c + \sum_{k=1}^p A_K Y_{t-k} + \Gamma X_t + \varepsilon_t$$

$$, \quad \varepsilon_t \sim \mathcal{N}(0, \Sigma),$$

	(1) Dependent Variable $L_{W,t}$	(2) Dependent Variable $L_{O,t}$
$S_W.lag5$	0.083**(0.025)	$S_O.lag5$ -0.040(0.003)
$S_W.lag6$	0.066**(0.023)	$S_O.lag6$ 0.812(0.531)
$N_W.lag1$	1.518**(0.481)	$N_O.lag1$ 1.044*(0.575)
$N_W.lag3$	1.617**(0.534)	$N_O.lag3$ 0.733(0.632)
$SE_W.lag2$	0.025*(0.009)	$SE_W.lag2$ -0.002(0.009)
$SE_O.lag4$	-0.010*(0.004)	$SE_O.lag4$ 0.007*(0.004)
Control	Yes	Yes
Observations	134	134
Adjusted R ²	0.728	0.740

Notes. Robust standard errors are in the parentheses. *** p < 0.005; ** p < 0.01; * p < 0.05.

Next: GNN Forecasting Results: Impact of Social Media Features

Model	MAE	RMSE	MAPE	sMAPE
LSTM	226.97	419.74	15.3%	15.5%
NHiTS	225.66	409.38	16.0%	15.2%
NBEATS	221.72	409.13	14.8%	14.8%
LSTM + Burst + GATED GNN (with Social Media)	182.19	336.90	12.7%	13.0%

NEW DESIGN is the best model across all 4 metrics

Improvement Over Best Baseline (NBEATS)

MAE

221.72 → 182.19

▼ **17.8% improvement**

RMSE

409.13 → 336.9

▼ **17.7% improvement**

MAPE

14.8 → 12.7

▼ **14.2% improvement**

sMAPE

14.8 → 13

▼ **12.2% improvement**

Social media NLP signals (52,866 Reddit shortage posts) provide leading indicator advantage, consistently outperforming all Rx-data-only baselines across every metric

PART 2 — Medical RAG: Problem & Motivation

Context: Refractive Surgery Post-Operative Care (LASIK · PRK · SMILE)

Physicians frequently answer the same recovery, complication & procedure questions. General LLMs hallucinate specifics or cite discontinued techniques based on the limitation of model

Field-Specific

Grounded in AAO
Preferred
Practice Patterns
(PPP 2024) and
related notebook

Accuracy

- No hallucinated
- Zero dangerous responses (score=1)

Empathy

- Patient-appropriate language
- Compassionate medical persona

Traceability

Source attribution to
specific
guideline & book

Solution: RAG + LLM → traceable, safe patient answers

PART 2 — Medical Document Dataset & Preprocessing

Dataset Statistics

800 Total medical documents (CHINESE)

784 PDF files (ophthalmology textbooks + PPP)

16 Word docs (clinical protocols)

2.27M Total characters

37 → 800 Large PDFs expanded (10-pg chunks)

4,119 Final text chunks (avg 45/doc)

800 chars Chunk size · 100-char overlap

Chunking Pipeline

PDF Splitting

Files >30 pages → 10-page chunks, 1-page overlap. Original 37 large files → 800 processable documents (<7MB each)

Table Extraction

PyMuPDF table detection → Markdown format + metadata. Captures surgical outcome tables & clinical comparison data

Image/OCR Extraction

Bounding box coordinates + OCR for embedded text. Anatomical diagrams stored as searchable image-caption metadata

Recursive Chunking

800-char chunks, 100-char overlap into 4,119 chunks. Metadata: source doc, page, content type, section hierarchy

PART 2 — Two-Stage RAG Architecture

Stage 1: Retrieval (Semantic Search)

Patient Query q

BGE Embeddings: BAAI/bge-base-zh-v1.5
12 layers · 768-dim · 512 max tokens

Mean pooling: $e = \text{mean}([h_1, \dots, h_n])$

ChromaDB HNSW indexing
Cosine similarity(q, d) = $q \cdot d / (||q|| \cdot ||d||)$

Retrieve top- $k=2$ docs where $\text{sim} > \tau=0.5$
(0.011s on A10G GPU — 100× CPU speedup)

Stage 2: Generation (Conditioned LLM)

Prompt = [System Role: Compassionate
Ophthalmologist]
+ [Retrieved Medical References D]
+ [Patient Question q]

DeepSeek-R1:1.5b via Ollama (local)
1.5B params · 934MB · Temperature $T=0.7$

Response r : 150–200 words
Evidence-based · Source-attributed

Hyperparameters optimized on 10 held-out dev questions: top- $k=2$ · $\tau=0.5$ · 300 chars/chunk · $T=0.7$
validated before final 25-question evaluation

PART 2 — Physician Evaluation Framework

Test Set: 25 Patient Questions (Ophthalmologist-Curated)

1 General procedure information

LASIK/PRK/SMILE basics, how it works

2 Eligibility criteria

Candidacy requirements, contraindications

3 Safety & complications

Risks, side-effects, risk quantification

4 Recovery & outcomes

Timelines, activity restrictions, healing

5 Procedure comparisons

LASIK vs PRK vs SMILE trade-offs

Controlled Evaluation Design

Condition A: Without RAG (Baseline)

LLM uses only parametric (training) knowledge — represents current common practice (patients using general-purpose LLMs)

Condition B: With RAG

LLM conditioned on top-k=2 retrieved AAO medical documents — identical system prompt, only difference is knowledge grounding

Controlled Variables

Same system prompt (compassionate ophthalmologist persona)
· Same 25 questions · Physician blinded to condition during scoring

5-Point Physician Scoring Rubric

1=Incorrect/dangerous · 2=Poor · 3=Adequate · 4=Good · 5=Accurate & complete; based on factual correctness, completeness, safety & appropriateness

PART 2 — EXAMPLES

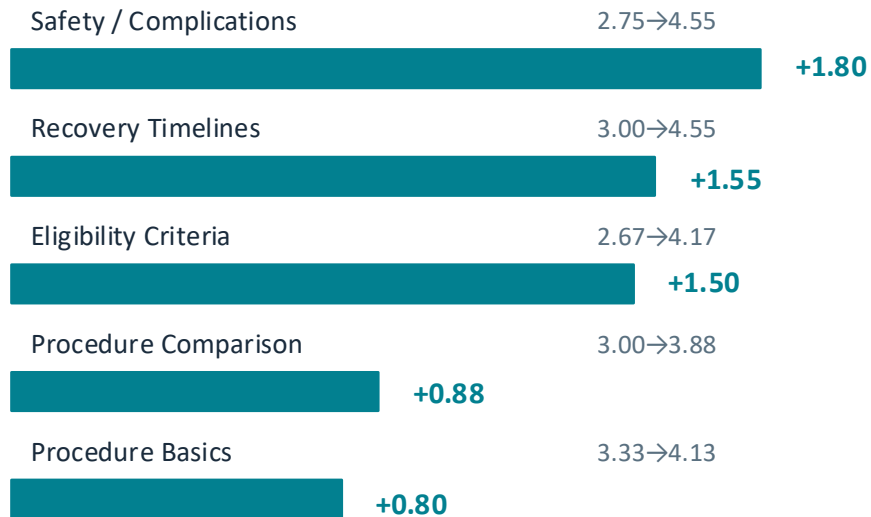
Question	Answer_NoRAG	Score_NoRAG	Answer_WithRAG	Score_WithRAG
1. What is refractive surgery?	Refractive surgery is a specialized eye care procedure aimed at correcting vision by altering the shape of the eye's lens. It targets areas where light needs to be properly focused, often requiring significant restructuring or removal of tissue around prescriptions. This allows clear lenses to focus on all parts of the eye, enhancing vision without heavy use in sensitive areas. Typically done by optometrists, it can improve conditions like myopia, hyperopia, or presbyopia. It's recommended for those with complex prescriptions or limited near vision, providing a way to maintain clear vision while minimizing discomfort.	3	**Refractive Surgery Explained** **What is Refraction Surgery?* Refraction surgery involves altering the shape of an eye to correct vision issues like myopia (nearsightedness), hyperopia (farsightedness), or astigmatism. It uses advanced technologies like corneal photocoherence to create artificial lenses. **Risks and Considerations:** **Side Effects:** Some patients may experience discomfort, nausea, or difficulty concentrating during surgery. **Post-Surgical Concerns:** If vision remains unclear after the procedure, further testing might be recommended. They'll need clear glasses or contact lenses tailored to their needs. **Types of Refraction Surgery:** **Keratoconstrictive Surgery:** Typically used for mild myopia or hyperopia. **Kerato-praxia (Topical/Regional Anesthesia):** May provide a more comfortable experience by reducing discomfort but may require local anesthesia. **What to Expect:** After surgery, patients may have difficulty concentrating. It's essential to guide them through the recovery process and remind them to consult their eye care physician for personalized advice. Letting your patient know that refractive surgery is a treatment option for vision correction can empower them.	4
2. Can I have both eyes treated at the same time?	Yes, certain ophthalmological treatments can indeed be performed on both eyes simultaneously. For example, laser therapy for glaucoma and photorefractive surgery for cataracts often involve treating both eyes together to reduce the impact of these conditions. However, most treatments require patient cooperation due to complexity or extensive procedures, necessitating a coordinated approach with a healthcare provider.	4	Yes, you can have both eyes treated simultaneously, but the treatment may need to focus on specific areas of each eye that don't interfere with others. For instance, if an issue in the cornea or lens is resolved together with those affecting only one eye, they can be treated at the same time. However, defects in the retina and optic nerve require separate attention for both eyes due to their interconnected nature. It's generally better to treat each eye separately unless combined strategies are safe and effective. Always consult a professional eye doctor for personalized advice.	5
3. What are the most common complications?	The most common complications of the eye include **cataracts** , which are a significant issue due to their frequent occurrence; **glaucoma** , another prevalent condition characterized by fluid retention inside the lens; **astigmatism** (also known as hyperopia), the most common form; **second degree of maculae damage** , responsible for vision loss later in life; **inherited or inherited eye conditions** such as age-related macular degeneration and cataracts due to genetic mutations; and **age-related changes in vision** , where the eyesight decreases with each year passing.	2	The most common complications of corneal refractive issues include: **Dry Eyes**: Inadequate fluids entering the eye due to dryness or excessive sweating. **Scarring (Edema)**: From sunburns, allergies, or infections, leading to redness and sensitivity in the area. **Dehydration**: Inadequate fluids moving within the eye, often caused by sunlight or excessive sweating. **Lubrication Issues**: From skin scratches or infections that affect fluid movement. **Cataract Formation/Changes**: Development of cataracts due to pressure on the corneum surfaces. **Acute Corneal Damage (Keratoopathy)**: Pressure on the back surfaces causing vision issues and discoloration. **Prolonged Use or Injury**: Causing severe changes in corneal structure, often leading to cataracts or other complications. **Atrus/Atoria**: Slowed eye movement, impacting reading and life functions. **Anomalous Corneal Surface**: Affects cataract formation and may lead to sensitivity in the back surfaces. **Pseudorhabdophenoid**: Commonly associated with genetic mutations affecting the epidermis or cornea. **Dry Eyes in Others**: In rare cases, even in patients without eye problems, dryness can manifest elsewhere. **Lacrimofacial Nucleus Mutation**: A genetic condition that may contribute to corneal changes. Additional rare complications include: **Atrus/Atoria**: Slowed eye movement, impacting life functions. **Cataract Formation/Changes** (rare). **Chromophore Dysfunction**: Leading to cataracts and vision changes. **Acute Choriorretia Syndrome**: Severe back surface pressure causing severe eye damage. These conditions often stem from genetic mutations or structural abnormalities in the cornea. Always consult an ophthalmologist for personalized advice.	5

Part 2 — Physician Scoring Results

3.04 → 4.36 Mean Physician Score **(+1.32 pts · +43.4%)** | Median: 3.0 → 4.0 | Range: [2,4]
→ [3,5] | Zero dangerous responses (1/5) in either condition

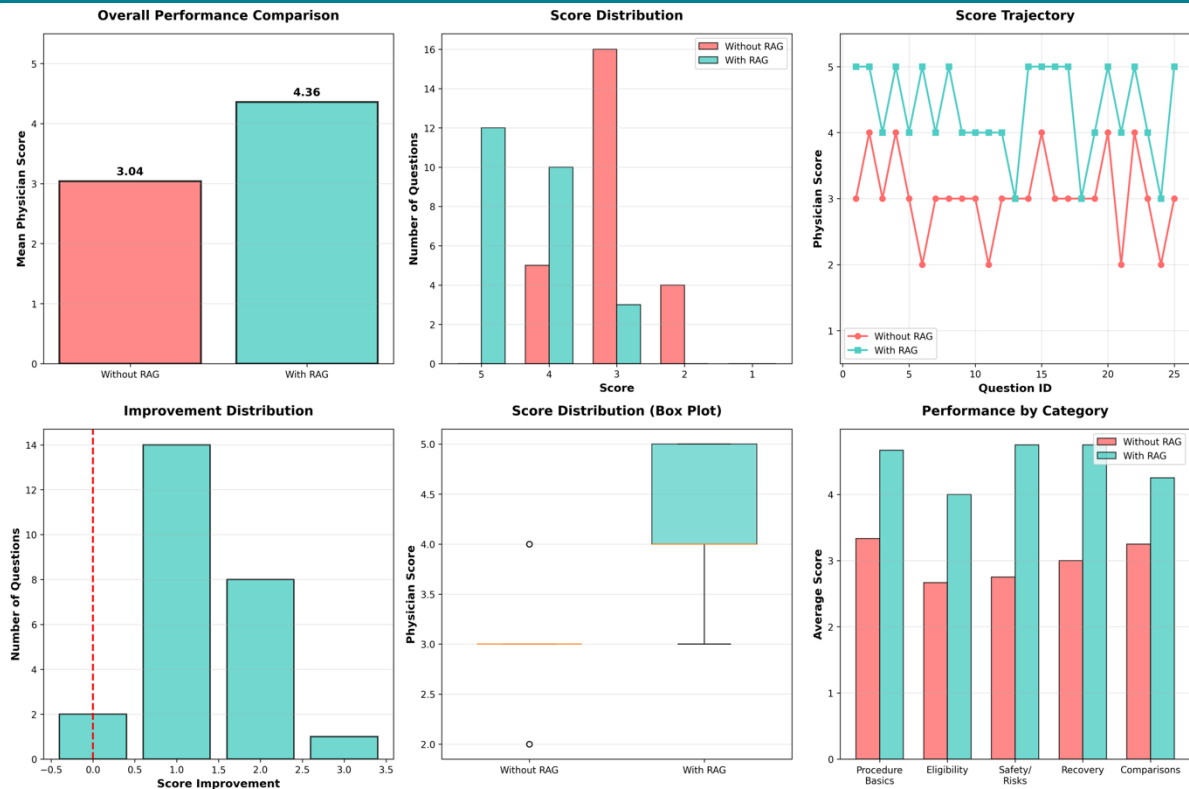
Score Level	Without RAG	With RAG
Excellent (5)	0 questions	12 questions ▲
Good (4)	6 questions	10 questions
Adequate (3)	14 questions	3 questions
Poor (2)	5 questions	0 questions ▲
Dangerous (1)	0 questions	0 questions

Performance by Question Category



RAG had greatest benefit for questions requiring specific clinical criteria & risk quantification

PART 2 — Visual Analysis: 6-Chart Performance Dashboard



① Overall Performance

② Score Distribution

③ Score Trajectory

④ Improvement Distribution

⑤ Box Plot Comparison

⑥ Performance by Category



Thank you for attention!